Hemaang Sood

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Skills

AI, LLMs & Machine Learning
PyTorch, TensorFlow, OpenAI API, Hugging Face Transformers, LangChain, DSPy, GEPA, RAG, OpenCV, OpenAI Gym, XGBoost.

Programming Languages
Python, JavaScript, Go, Java, Kotlin, R.

Backend & Web Development
FastAPI, Django, Spring Boot, NodeJS, React.js, NextJS, Remix, Redux, Tailwind CSS, Three.js, React Three Fiber.

Data, Retrieval & Databases
Pandas, NumPy, Matplotlib, Plotly, SQL, FAISS, Qdrant, PostgreSQL, MongoDB, Supabase, Redis.

Cloud, DevOps & Tools
AWS (EC2, S3), Docker, Docker-in-Docker, Git, GitHub, Nginx, Postman, Poetry.

Testing & CI/CD
pytest, GitLab CI/CD.

Experience

Synechron
Jr Associate

- Completed introductory sessions on **Capital Markets** and **Commodities**, developing foundational understanding of financial systems and trading concepts.
- Built a full-stack project for an **Amusement Park Management System** using **MERN** with **Vite.js** and **Tailwind CSS**, focusing on modular UI design, API integration, and database connectivity.
- Built a backend server project for an **Insurance Management System** using Springboot, focusing scalability security and modularity.

Bharat Electronics Limited (BEL)
Intern

- Developed an Authentication and Authorization module for DRDO as part of the internship as part of a secure intranet website.
- Worked on a Large Language Model (LLM) interface to locally process and extract insights from sensitive documents using **RAG**. Integrated a **voice input pipeline** using Vosk to transcribe user speech into queries, enhancing accessibility and usability.

Education

University of Waterloo — Jan 2026 – April 2027
Electrical & Computer Engineering, MEng — GPA - 3.9 / 4.0

BMS Institute of Technology and Management (BMSIT&M) — 2021 – 2025
Bachelor of Engineering in Computer Science and Engineering — CGPA - 8.9 / 10.0

Conference Publications & Presentations

- Hemaang Sood et al., "Improving Face Recognition for Smart Home Security: A Study of ResNet, Siamese Networks, and Attention Mechanisms" Presented at IEEE COMPSIF 2025.

Personal Projects

Virtual POCT — AI-Powered Quality Control Assistant: *Individual*

- Built a full-stack AI-powered quality control platform for Point-of-Care Testing (POCT), enabling healthcare operators to troubleshoot devices through a conversational assistant with auditable QC workflows.
- Designed a streaming WebSocket architecture using FastAPI and LangChain to deliver real-time LLM responses while supporting concurrent multi-session conversations.
- Implemented a deterministic quality control evaluation engine that combines operator input with database context to classify QC scenarios and automate compliance actions.
- Developed a Dockerized microservice architecture with React, FastAPI, PostgreSQL, and Nginx, including JWT authentication, role-based access control, and persistent session history.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, SQLAlchemy, Docker, Nginx, WebSockets, LangChain, Google Gemini API, JWT.

ECE 699 Research Project — Multi-Agent LLM Optimization with DSPy and GEPA: *Group*

- Conducting research on optimizing **multi-agent LLM systems** using **DSPy** and **GEPA**, focusing on systematic optimization of agent programs and performance, specifically **automatic prompt optimization in multi-agent systems using genetic algorithms**.
- Implementing and empirically evaluating optimization approaches for **multi-agent LLM workflows**.

Facial and Voice Recognition-Based Door Lock System: *Group*

- Achieved 98.7% accuracy in preventing unauthorized access by integrating facial and speech recognition algorithms for secure authentication.
- Processed and augmented the YaleFace dataset, utilizing a custom YOLO model to extract, align, and normalize facial features for improved model training.
- Implemented a Siamese Network to optimize feature comparison, enhancing the precision and robustness of facial recognition for reliable access control.
- Built Backend in Django and made a NGINX reverse proxy.
- Published paper in **IEEE Compsif 2025** for the same.
- **Technologies and Tools:** OpenCV, PyTorch, Google Speech API, RESTful APIs, Python, Raspberry Pi, Hybrid Neural Networks, Docker, React, PostgreSQL, Tailwind, Shad CN, AWS EC2, Django, NGINX.

Routed: Travel Partner Recommendation System *Group*

- Designed and developed a microservices-based platform to match travelers with compatible trip partners using trip overlap, preferences, and geospatial proximity.
- Implemented a FastAPI backend with JWT authentication, trip management APIs, and a matching system powered by PostgreSQL and PostGIS for spatial queries.
- Built an event-driven notification pipeline using Kafka and a Go-based worker service that processes match events and stores notifications in MongoDB.
- Built an Kotlin microservice with coroutines to extract semantic embeddings from bios.
- **Technologies Used:** FastAPI, React, TypeScript, PostgreSQL, PostGIS, Kafka, MongoDB, Docker, Go, SQLAlchemy.

Resume Builder: *Individual*

- Built an AI-powered resume optimization tool that parses LaTeX resumes, tailors content to job descriptions, and generates polished, job-specific resume output.
- Agent utilizes multi-step process to parse optimize and output the the optimized resume.
- **Technologies Used:** Python, OpenAI API, LaTeX, SQLite, Jinja2, Rich, pdflatex, python-dotenv

MERN-Stack Based Chat App: *Individual*

- Created Highly secure end-to-end encrypted chat app with 256 bit AES-256 encryption.
- **Technologies Used:** MongoDB, Express.js, React.js, Node.js, Tailwind, Next-Auth, Cryptography.

Stock Analysis using Machine Learning: *Individual*

- Collected, cleaned, and transformed historical stock data for 25 stocks, integrating 10 global economic factors to enhance model performance and generate actionable insights.
- **Technologies and Techniques:** LSTM, GRU, CNN, Reinforcement Learning, Hybrid Neural Networks, PyTorch, Python, AWS (EC2, S3), Pattern Recognition, Data Preprocessing, Feature Engineering, Statistical Analysis.

Reinforcement Learning for Snake Game: *Individual*

- Created Convolutional Neural network that learnt to complete snake game within 45 minutes of training.
- **Technologies Used:** Python, TensorFlow, Deep Q-Learning, RL, OpenAI Gym, PyGame, CNN, NumPy, Matplotlib.